BITS Pilani — Work Integrated Learning Programmes (WILP)

M.Tech (AIML) | Machine Learning (AIMLZG565) — Assignment 2

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1. Mandatory Submission Links

1. GitHub Repository Link: https://github.com/AdiS-2105/ml-assignment-2

Contains: Complete source code, `requirements.txt`, comprehensive `README.md`, `test_data.csv`, Jupyter notebook `ML_Assignment_2_Notebook.ipynb`, and serialized pre-trained models in `model/`.

2. Live Streamlit App Link: https://2026aj05162-ml-assignment-2.streamlit.app

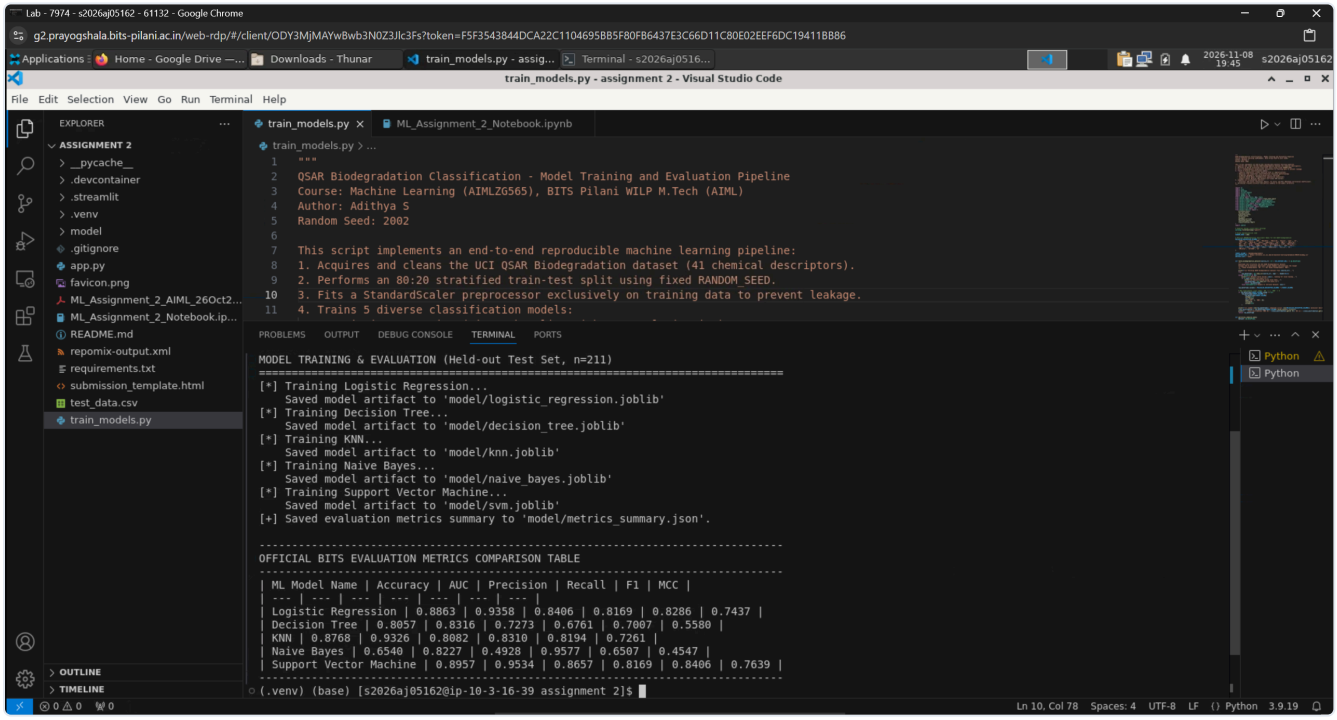
Interactive web application deployed on Streamlit Community Cloud (Free Tier) with dataset uploader, model dropdown, live evaluation metrics, confusion matrix, and 5-model comparative benchmark.

2. BITS Virtual Lab Execution Proof Screenshot [1 Mark]

The entire assignment modeling pipeline was executed within the **BITS Virtual Lab** environment (`g2.prayogshala.bits-pilani.ac.in`) under student account **`s2026aj05162`** (terminal: `[s2026aj05162@ip-10-3-16-39 assignment 2]$`).

Figure 2.1 — Model Pipeline Training & Evaluation Benchmark in Terminal BITS Virtual Lab Proof (Primary)

Execution of `train_models.py` in VS Code terminal on BITS Virtual Lab (user: `s2026aj05162`). Demonstrates training all 5 classifiers, artifact serialization into `model/`, and printing the official 5-model evaluation metrics table.



2. BITS Virtual Lab Execution Proof (Continued)

Figure 2.2 — Exploratory Data Analysis & Feature Importance in Jupyter Notebook

BITS Virtual Lab Notebook

Execution of `ML_Assignment_2_Notebook.ipynb` in BITS Virtual Lab environment (Python 3.12.7 kernel). Visualizes Top Molecular Descriptors by Standardized Log-Odds (Logistic Regression) and Top 10 Features by Gini Impurity Reduction (Decision Tree).

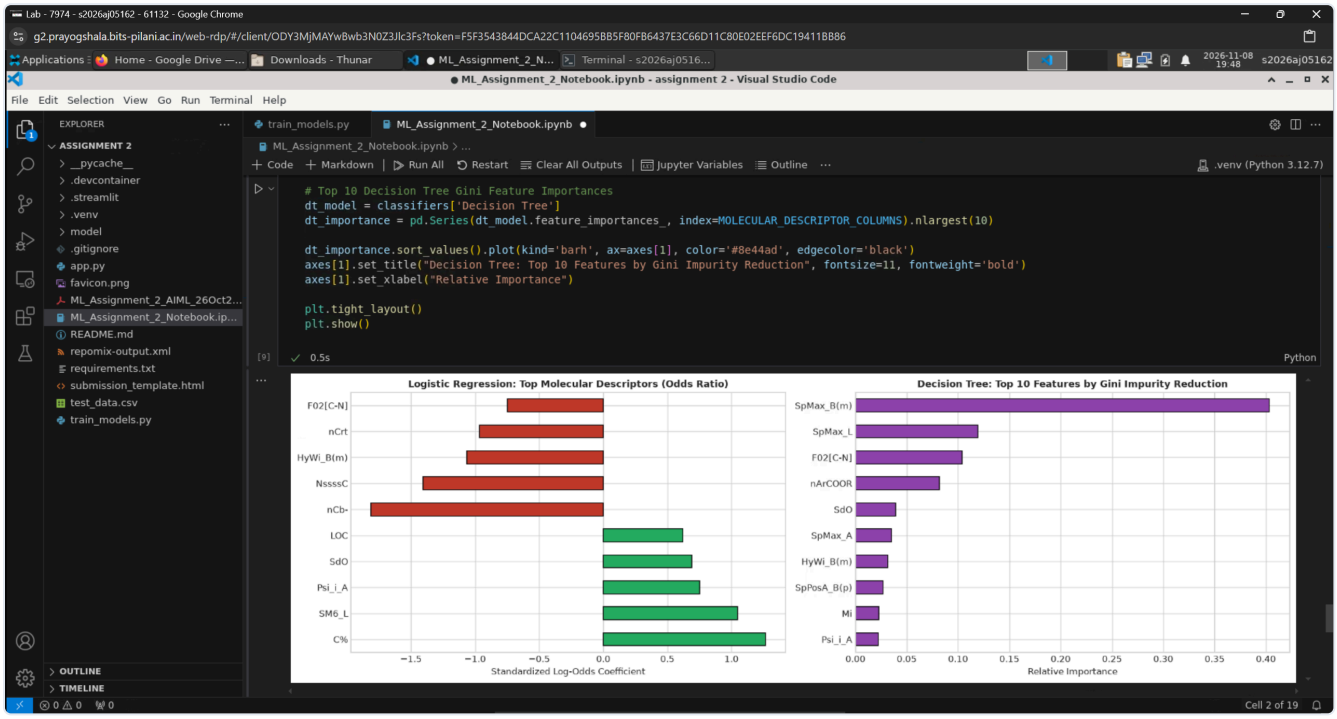
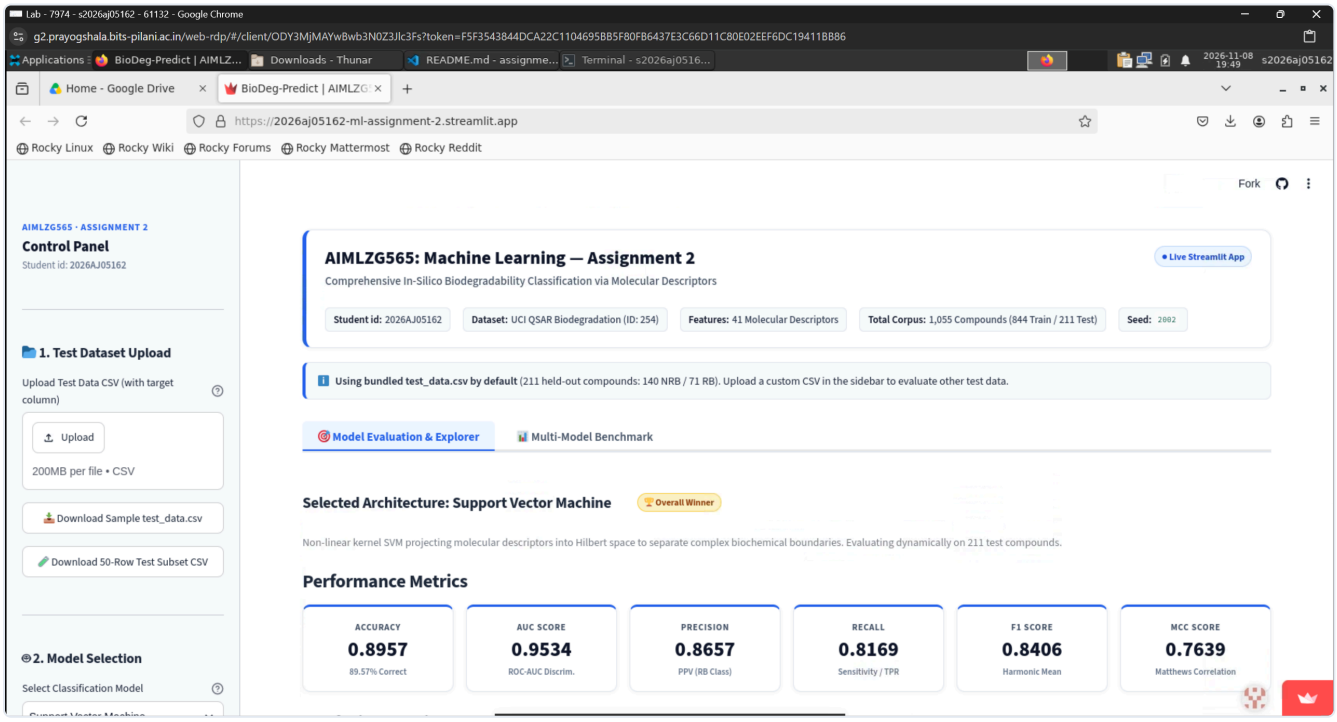


Figure 2.3 — Live Streamlit Cloud Deployment Verified in BITS Virtual Lab Browser

Streamlit Cloud Deployment

The deployed Streamlit Cloud application (`https://2026aj05162-ml-assignment-2.streamlit.app`) verified inside Google Chrome on BITS Virtual Lab. Displays student ID `2026AJ05162`, high-contrast light theme, bundled `test_data.csv` status, and live Support Vector Machine metrics.



3. GitHub README.md Content

a. Problem Statement

Evaluating whether organic chemicals readily biodegrade in aerobic aquatic environments is an indispensable safety requirement under international regulations (e.g., EU REACH regulation and OECD 301 testing guidelines). Laboratory degradation assays require prolonged 28-day microbial incubation tests that are expensive, labor-intensive, and impractical for screening large chemical spaces.

Quantitative Structure-Activity Relationship (**QSAR**) modelling provides a rapid in-silico alternative by predicting experimental biodegradability directly from computed molecular descriptors. In this project, we implement and benchmark five distinct machine learning classification architectures to predict whether a compound is **Ready Biodegradable (RB, 1)** or **Not Ready Biodegradable (NRB, 0)** based on 41 molecular descriptors.

b. Dataset Description [1 Mark]

The study employs the benchmark **UCI QSAR Biodegradation Dataset** (ID: 254), compiled by the Milano Chemometrics and QSAR Research Group.

- **Total Instances:** 1,055 organic compounds (≥ 500 rows requirement satisfied).
- **Total Features:** 41 molecular descriptors + 1 target variable (≥ 12 features requirement satisfied).
- **Missing / Null Values:** 0 across all 41 continuous and discrete descriptors.
- **Target Class Distribution:** 699 Not Ready Biodegradable (NRB, 0) and 356 Ready Biodegradable (RB, 1) (~2:1 ratio).
- **Stratified Train-Test Split (80:20):** 844 training compounds and 211 held-out test compounds using fixed `RANDOM_SEED = 2002`. The held-out test set is saved as `test_data.csv`.
- **Leak-Free Preprocessing:** `StandardScaler` was fitted exclusively on the 80% training set and saved as `model/preprocessor.joblib`. Test samples are transformed without data leakage.

c. GitHub Repository Structure [1 Mark]

Repository Link: https://github.com/AdiS-2105/ml-assignment-2

```
project-folder/
├── app.py                          # Interactive Streamlit application
├── requirements.txt                # Pinned production dependencies
├── README.md                       # Full documentation & rubric guide
├── test_data.csv                   # Held-out test set (n=211 compounds)
├── ML_Assignment_2_Notebook.ipynb   # EDA & training notebook
├── train_models.py                  # Standalone reproducible training script
└── model/                           # Pre-trained model artifacts & preprocessor
    ├── logistic_regression.joblib, decision_tree.joblib, knn.joblib
    ├── naive_bayes.joblib, svm.joblib, preprocessor.joblib
    └── metrics_summary.json
```

d. Models Used & Evaluation Metrics Comparison Table [5 Marks]

Performance of all five models evaluated on the held-out test set (n = 211; 140 NRB, 71 RB):

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.8863	0.9358	0.8406	0.8169	0.8286	0.7437
Decision Tree	0.8057	0.8316	0.7273	0.6761	0.7007	0.5580
KNN	0.8768	0.9326	0.8082	0.8310	0.8194	0.7261
Naive Bayes	0.6540	0.8227	0.4928	**0.9577**	0.6507	0.4547
Support Vector Machine 🏆	**0.8957**	**0.9534**	**0.8657**	0.8169	**0.8406**	**0.7639**

e. Observations on Model Performance [3 Marks]

ML Model Name	Observation about model performance
Logistic Regression	**Strong Linear Baseline:** Achieves 88.63% Accuracy and 0.9358 AUC. The L2 penalty effectively regularizes collinear descriptors. Higher oxygen counts (`nO`) and ester bonds (`nArCOOR`) strongly increase the log-odds of biodegradation, matching biochemical oxidation mechanisms.
Decision Tree	**Axis-Aligned Stepped Partitions:** Yields 80.57% Accuracy and 0.5580 MCC with pre-pruning (`max_depth=5`). Orthogonal decision hyperplanes struggle to approximate smooth diagonal interactions across continuous molecular manifolds.
KNN	**Congeneric Neighborhood Clustering:** Achieves 87.68% Accuracy and 0.9326 AUC. Effectively leverages the chemical similarity principle in 41D standardized metric space.
Naive Bayes	**High Sensitivity / Independence Violation:** Achieves the highest Recall across all models (**95.77%**), correctly detecting 68 out of 71 biodegradable compounds. However, strong descriptor inter-correlations violate conditional independence, causing 70 false positives. Serves as an effective conservative screening filter.
Support Vector Machine	**Optimal Non-Linear Boundary:** Top performer in Accuracy (**89.57%**), AUC (**0.9534**), Precision (**86.57%**), F1 (**0.8406**), and MCC (**0.7639**). The RBF kernel maps 41D space into an infinite-dimensional Hilbert space where complex bioactivity surfaces become linearly separable with maximum margin.
Overall, Winner for your dataset?	**Support Vector Machine (RBF Kernel) 🏆: The definitive winner. On moderate sample sizes (n = 844) with high feature dimensionality (d = 41), kernel SVM optimizes structural risk minimization, generating the fewest false positives (only 9/140) and highest correlation (MCC = 0.7639).**

f. Interactive Streamlit Web Application Features [4 Marks]

- **Dataset Upload (CSV):** Supports custom test CSV uploads with automatic column validation, along with built-in default loading of `test_data.csv` and download triggers.
- **Model Selection:** Interactive dropdown allowing seamless evaluation switching across all 5 models.
- **Evaluation Metrics Display:** Custom high-contrast metric cards for Accuracy, AUC, Precision, Recall, F1, and MCC.
- **Diagnostic Visualizations:** High-resolution Seaborn confusion matrix heatmap with exact cell counts alongside the complete per-class classification report.
- **Multi-Model Benchmark & Winner Showcase:** Interactive side-by-side comparison table, MCC/Accuracy radar comparison, and overall winner badge.
- **Zero Retraining Guarantee:** Strict evaluation-only pipeline leveraging pre-trained artifacts in `model/` without data leakage.

g. Anti-Plagiarism & Academic Integrity Statement

I declare that this assignment submission is my own original work. The models, evaluation metrics, pipeline scripts, and interactive Streamlit application were built and validated in compliance with the BITS Pilani WILP academic integrity guidelines. All experiments use reproducible fixed random seed `2002`.

Submitted by: Adithya S (ID: **2026AJ05162**) | BITS Pilani Work Integrated Learning Programmes (WILP) | Course: Machine Learning (AIMLZG565) — Assignment 2